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**A REVIEW ON ETHICAL ISSUES OF ANIMAL
EXPERIMENTATION IN INDIA-Gudivada Mani*,
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A REVIEW ON ETHICAL ISSUES OF ANIMAL EXPERIMENTATION IN INDIA

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ABSTRACT:

One of the pivotal issues in scientific research is to consider ethics in animal experimentation. Large number of scientific advances of the past century and our understanding of various diseases are contributions of Animal

research. Issues raised by research involving animals have aroused intense debate. Discussion on the subject is often portrayed as being essentially between two positions that are either 'for' or 'against' the use of animals. This is unhelpful, since the matter itself is complex, as are the many views that surround it. One group favours the use of animals in research and emphasizes the scientific and medical benefits that have arisen. Others are less focused on the scientific issues, and more concerned with the fundamental ethical question of whether it is right for humans to subject sentient animals to procedures that may cause them pain and suffering, and from which they will not benefit. This review discusses the need for animal experimentation and ethical issues related to animal research.

Key words: CPCSEA, Animal Research, Ethics, Experimentation

INTRODUCTION:

In the life of human beings animals have played a very crucial role. To understand several diseases, experimentation in animals has given us enormous information and knowledge. As there are no other alternatives to substitute biological system so far, Animal experimentation in biomedical research continues to remain important to understand the course of human diseases further for prevention and treatment [1]. The animal experimentation dates back to the period

when humans started to look for ways to prevent and cure ailments. Research on living animals has been practiced since ancient times. Animal experimentation is used as part of development mandatory requirement for testing drugs and medicines in pharmaceutical industry and also for disease diagnosis, teaching, and training. As alternatives to use of animals for pharmacology education and research, a number of computer simulation and other models have been recommended for use. As there are no other alternatives to substitute biological system so far, animal experimentation continues to remain crucial to a high proportion. Therefore, to improve the health of humans and animals, even today animals are used indispensably in research [2].

History of Animal Experimentation:

In many countries, animal experimentation has a long history. For example, Galen's work on animals where he performed vivi-section was severely opposed by the church [3, 4]. Vesalius also faced problems. Rene Descartes, the French philosopher of science and Mathematician was also a self styled Biologist. The remark made by Descartes when there was no use of anesthetics, that animals have no feelings and hence can be freely used for dissection affected Biology. There was no legislation to regulate such crude animal experimentation. Only in the nineteenth century some legislative measures to regulate research using animals was brought by England and USA, which became more frequent [5]. Other countries followed such practices. A running battle between scientists and government bodies was resulted as a consequence. Even in our country two extreme and mutually exclusive stands taken in the initial days. The Society for Prevention of Cruelty to Animals (SPCA) took strong exception to protocols followed by scientists even in leading laboratories. Either you are allowed to use animals for research by default or not permitted to use them. Scientists or activists took up adversarial positions. Some rationalization introduced in the use of animals for research because of negotiations among the stakeholders. Guidelines were framed which satisfied both the stakeholders i.e. animal activists and scientists but to different degrees. The Indian National Science Academy (INSA) took the early lead, in framing these reasonable guidelines through Prof PN Tandon in nineties. The administrative framework

for implementing these guidelines was also in place in the form of institutional animal ethics committees and the CPCSA rules. In early 21st century; when animal activists were replaced by working scientists as CPCSCA nominees, a quantum jump in research was observed. The guidelines were followed more faithfully by Scientists [6].

Animals used in Research:

For animal testing around the world, many different species of animals are used. Birds and fish are used less frequently, but mini-pigs, farm animals, frogs, mice, rabbit, guinea pig, sheep, albino rats, primates etc., are used most commonly. Mice is a common animal model. [7].

Reasons for using animals in research:

Because of evolutionary ties and common ancestry, mammals and all vertebrates have strong similarities. Mammals that look quite different from us share a large percentage of our genes; small and furry mice share 92% of our genes. Our closest living animal cousins, Chimpanzees share 98% of our human genes. Approximately 44% of our human genes are shared by fruit flies [8]. Since human testing is not ethical and legal, the genetic similarity of animals has allowed researchers to use them as human models. Some animals have biological similarities to humans that make them particularly good models for specific diseases, such as rabbits for atherosclerosis or monkeys for polio. While humans cannot be used in research for ethical reasons, variables (like diet, housing, clean air, humidity, temperature, and genetic makeup) which could not be controlled in human subjects, can be controlled in a mouse [9].

Most people believe that limited and regulated animal use is justified, in order to achieve medical progress that will save and improve lives. More than half the drugs used by vets were developed originally for human medicine; like that animal research benefits animals too. Some of the most important drugs and treatments that we use today are based on animal testing. Through the use of animals in research, hormones like Insulin, vaccines, antibiotics, and numerous cancer and HIV treatments were developed [10]. The mouse has become the surrogate for human disease, because of our ability to manipulate the genome of the laboratory mouse [11].

There are four main reasons for using animals in research:

- It helps to understand how organism function and develop. Although it is more academic in nature, basic research is intricately linked to applied or practical research [12].
- Animal models help researchers to understand the type of damage happens in the body and in the development of new
- Before testing on human patients, new therapeutic techniques and surgical procedures were developed and tested in
- To protect the safety of people, animals, and the environment, testing on animals is

Animal Experimentation in India – Ethics, Laws and Regulations:

Animal experimentation forms a big part of pharmaceutical research. To decide whether a particular drug should be tested on people or not, animal experiments help. Animal experiments eliminate some potential drugs as either ineffective or too dangerous to use on human beings. Before large scale clinical trials, a drug can be tested on a small human group, but before that it should pass the animal test [13]. So, animal testing is a primary step before going for clinical trials.

The ethical assessments related to the use of animals in research are wide-ranging. Many people believe that it is wrong to make animal suffer without reason; but some people think that there is some condition more than the interests of people outweigh the interests of an animal, this justifies animal suffering more than enough.

The general opinion is that animals have a moral status, and that our treatment of them should be subject to ethical considerations. Such views are reflected in the following positions [14]:

- An intrinsic value of the animals must be
- Interests of animals must be taken into consideration, because the animals are sentient creatures with the capacity to feel pain
- The use of animals in research is an expression of our attitudes and influences us as moral actors.

Humane animal experimentation techniques, the 3Rs – Replacement, Reduction, and Refinement are categorized in 1959, by William Russell, an intelligent young zoologist; and Rex Burch, a microbiologist, in their publication “The Principles of Humane Experimental Technique.” [15]. To this, the present day animal activists have added another “R”– Rehabilitation. Animal experimentation ethics did not emerge de novo. It evolved over centuries of philosophical traditions [16]. Only after the pressure from the animal protection groups and public, the regulatory bodies came into existence. These regulatory groups enacted laws to regulate the care and usage of laboratory animal models.

Ethical guidelines of animal testing [17,18]:

The following are global guidelines of animal testing (Table:1)

- CPCSEA Committee for the Purpose of Control and Supervision on Experiments on Animals
- CARE Community on Animal Research and Ethics
- GCUASR Guidelines For Care And Use Of Animals In Scientific Research
- NACLAR National Advisory Community for Laboratory Animal Research
- APA American Psychological Association

Even before the PCA Act came into existence, for the first time way back in 1957 in Mumbai, a centre was created in India, exclusively for Laboratory Animal Science. This centre was called Laboratory Animal Information Service (LAISC) with the funding support from UNESCO. Subsequently it was taken over by ICMR in 1959. To understand the problems that existed within those scientific fields in which live animals were used for experimental procedures, a survey was undertaken by the centre for more than 2 years. The centre

continues to be under ICMR and shifted to Hyderabad in 1976 and is now known as National Centre for Laboratory Animal Sciences [1].

Currently in India, experimentation on animals is covered under the provisions of Prevention of Cruelty to Animals Act, (PCA Act) 1960 and the Rules under the amended Act of 1998 and 2001. This is implemented through a committee called "Committee for the Purpose of Control & Supervision of Experiments on Animals (CPCSEA)." It is a statutory body which was established in 1964 under Section 15(1) of Chapter 4, of the PCA Act under the Ministry of Environment Forest & Climate Change. The CPCSEA provides guidelines for performing experiments on animals and maintenance of animal house [19].

Table 1: Different Community Work for the Ethical Guidelines for Animal

	CPCSEA	CARE	GCUASR	NACLAR	APA
Full Form	Committee for the Purpose of Control and Supervision on Experiments on Animals	Community on Animal Research and Ethics	Guidelines For Care And Use Of Animals In Scientific Research	National Advisory Community For Laboratory Animal Research	American Psychological Association
Guidelines Followed					
In Country/ Countries	India	Washington, DC	Indian National Science Academy(India)	Australia, Canada, New Zealand, US,	US
Goal	Guidelines is to promote the humane care of animals used in biomedical and behavioural research and testing with the	Important parts of these endeavours are teaching and research on the behaviour of nonhuman animals, which	The interest of society and the welfare of animals.	Guidelines set out the responsibilities of all the parties involved in the care and use of animals for scientific	The acquisition, care, housing, use, and disposition of nonhuman animals in research must be in compliance with

basic objective of	contribute to the	purposes, in	applicable federal,
providing	understanding of	accordance with	state, and local,
specifications that	basic principles	widely accepted	laws and
will enhance	underlying	scientific, ethical	regulations,
animal well-	behaviour and to	and legal	institutional
being, quality in	advancing the	principles	policies, and with
the pursuit of	welfare of both		international
advancement of	human and		conventions
biological			
knowledge that is	non human		to which the
relevant to	animals.		United States is
humans and			a party.
animals.			

The registration of animal house is mandatory with CPCSEA and is to be renewed every 3 years. At present, in India, there are 1723 facilities registered with CPCSEA for conducting experiments using animals. The Indian National Science Academy (INSA) and Indian Council of Medical Research (ICMR) have also formulated certain guidelines for care and use of animals in scientific research as well as in medical colleges, besides the rules and procedures laid down by the CPCSEA [20, 21].

The Pharmacy Council of India (PCI) issued a directive to all pharmacy schools in India to use CAL software in place of classroom animal experiments in 2003[22]. As an alternative to these tests involving animals, JIPMER, Pondicherry, has developed EX-PHARM Blank CD, which has been specially prepared as a 100% replacement to animals used in undergraduate courses in Medicine, Pharmacology, and Veterinary Science [23]. In 2011, the University Grants Commission banned dissection or experimentation on live animals in Zoology and life science courses [24].

Prohibition of animal testing for household products manufactured in India was established in April 2016, by the Union Minister Menaka Gandhi along with the Indian Ministry of Health and Family Welfare. Similarly, cosmetic testing in animals was banned by European Union, Norway, and Israel [25]. However, the Chinese government requires animal testing on certain cosmetics that are produced domestically (such as hair coloring and sunscreen) and requires animal tests for all imported cosmetics.

Laws and regulatory practices of the ethics committees should be strictly enforced as animal testing is a necessary evil; it is indispensable and inevitable. It should be viewed in a broader perspective in the larger interest of a nation's progress [26].

Different Alternative Methods:

There are four methods of Alternatives to animal use in biomedical and behavioural research :

- **Substitution method:** includes substitution of cold-blooded for warm-blooded vertebrates, substitution of laboratory mammals for domestic or companion mammals, and reduction of pain or experimental insult;
- **Reduction method:** a reduction in the number of animals used, improved experimental design and statistical analyses of results, use of living systems, including in vitro cultures (of cells, tissues, and organs) embryos, invertebrates, microorganisms, and plants;
- **Replacement method:** Use of non-living systems by using chemical or physical systems; and

Computer simulation.

The needs of health research are growing currently and medical developments will continue to depend on animals in future also. There are several questions to be answered regarding diseases / conditions like Stem-cells, gene therapy which is in initial stages. Hunt for development of vaccine and therapies for AIDS and Malaria are not over yet. Diseases like Alzheimer's, Parkinson's, are at the forefront still, there is a lot to be done with regard to the development of more efficient prophylactic vaccines and curative drugs. In the recent times there has been increasing threat from emerging historical diseases like Leprosy, TB and new diseases such as SARS, Chikungunya, Dengue etc. The pandemic of coronavirus disease 2019 (COVID-19), caused by SARS-CoV-2, poses an extraordinary threat to global public health, socioeconomic stability, food security and other social goods [27, 28]. A number of new drugs and new generation vaccines have been developed by recombinant technology and many are in pipeline for combating the challenges posed from these diseases and many others.

Alternative methods and animal experiments are used in a complimentary manner both in basic and in application-oriented research. While individual molecular and cellular aspects of life processes are researched in vitro or in silico as much as possible, scientific work on animals is irreplaceable in order to further understanding of our complex relationships within the entire organism. We have the chance to significantly reduce the number of test animals and the stress caused to them, by carefully considering and selecting the most suitable methods [29].

Conclusion:

This review highlights the history of animal experimentation, animals used in research, reasons for using animals in research, common guiding documents on the ethical treatment of animals and alternative methods of animal experimentation. The use of animals in research is indispensable in several fields hence the animal use in research may be ethical in some way where alternative methods are not known or very expensive. The different methods which are alternatively used are sometimes beneficial sometimes not as seen in animal use too. Hence if experiment were performing on animal should be ethical as given in the guidelines for animal. Primary ethical considerations are that sampling and processing activities must adhere to relevant legislation and should be approved by at least an institutional AEC, the welfare of the study animals should be of primary concern and all efforts should be made to prevent or reduce pain, suffering and distress, sample sizes should be kept to the minimum, but sufficient to achieve the objectives of the study, the conservation status of the target organism should be considered, standard and accepted procedures should be followed. The review will provide a useful resource for journal editorial committees developing ethical guidelines for publication and for potential investigators designing and conducting research on animals.

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